



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY MOLECULAR SPECTROSCOPY				
Type of Course	Major Elective (SPECIALIZATION II: PHOTONICS)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY4CJ205- Modern Physics				
Course Summary	Introductory Molecular Spectroscopy provides a comprehensive overview of the principles governing the interaction between light and molecules. Students delve into spectroscopic techniques such as infrared, ultraviolet-visible, and nuclear magnetic resonance spectroscopy, gaining insights into molecular structure, dynamics, and interactions.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Gain basic knowledge on electromagnetic spectrum, spectral lines and diverse branches in spectroscopy	U	C	Viva Voce/ Seminar / Quiz
CO2	Gain theoretical know-how on rotational spectrum of diatomic and polyatomic molecules	An	P	Practical Assignment / Group Discussion
CO3	Gain theoretical know-how on vibrational spectrum of diatomic and polyatomic molecules	Ap	P	Seminar Presentation /

				Group Tutorial Work
CO4	Gain theoretical know-how on vibrating rotators and Born-Oppenheimer approximation	An	P	Instructor-created exams / Home Assignments
CO5	Gain theoretical know-how on Raman spectrum	Ap	M	Viva Voce
CO6	Gain practical knowledge on emission and absorption spectra	C	M	Group Discussion/ Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Introduction to Spectroscopy		5	8
	1	Quantization of energy, Regions of spectrum	2	
	2	representation of spectra, Basic elements to practical spectroscopy	1	
	3	signal-to-noise ratio	1	
	4	width and intensity of spectral lines	1	
	Sections 1.1-1.7 of Book 1			
II	Microwave Spectroscopy		11	16
	5	Rotation of molecules, Rotational spectra	2	
	6	Rigid diatomic molecules, Intensities of spectral lines	2	
	7	Effect of isotopic substitution, Non-rigid rotator	2	
	8	The spectrum of non-rigid rotator	2	
	9	Polyatomic molecules – linear, symmetric and asymmetric top molecules, Stark effect	3	
	Sections 2.1-2.4 of Book 2			
III	Infra-red Spectroscopy		16	23

	10	Vibrating diatomic molecule - Energy of a diatomic molecule	2	
	11	Simple harmonic oscillator, Anharmonic oscillator	3	
	12	Diatomic vibrating rotator,	2	
	13	Vibration-Rotation spectrum of CO, Born – Oppenheimer approximation	3	
	14	Effect of Breakdown of Born Oppenheimer approximation	2	
	15	Vibration of polyatomic molecules	2	
	16	Influence of rotation on the spectra of polyatomic molecules, Analysis by infra-red techniques	2	
	Sections 3.1-3.7 of Book 1			
IV	Raman Spectroscopy		16	23
	17	Quantum and Classical approach towards Raman effect	3	
	18	Pure rotational Raman spectra of linear, Symmetric top and spherical top molecules	3	
	19	Vibrational Raman spectra, Rule of mutual exclusion	3	
	20	Overtone and Combination vibrations, Rotational fine structure	2	
	21	Polarization of light and Raman effect	2	
	22	Raman & IR Spectroscopy in structure determination, Instrumentation	3	
	Sections 4.1-4.6 of Book 1			
V	Open Ended Module			
	Electronic Spectroscopy of molecules Electronic spectra of diatomic molecules, Vibrational coarse structure: Progressions, Intensity of vibrational-electronic spectra: Franck Condon Principle, Dissociation Energy and Dissociation Products, Rotational fine structure of Electronic-Vibration transitions, Fortrat Diagram, Pre-dissociation			
Books and References: 1.Fundamentals of Molecular Spectroscopy by C N Banwell, McGraw Hill (Book 1) 2. Molecular Structure & Spectroscopy by G Aruldas (Book 2) 3.Spectroscopy (volumes) by B P Straughan and S Walker 4.Introduction to Molecular Spectroscopy by G M Barrow, McGraw Hill				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	0	0	0	0	0	2	2	2	0	0	0	0
CO 2	2	0	2	0	0	0	2	2	2	0	0	0	0
CO 3	2	0	2	0	0	0	2	2	2	0	0	0	0
CO 4	2	0	2	0	0	0	2	2	2	0	0	0	0
CO 5	2	0	0	0	2	0	2	2	2	0	0	0	0
CO 6	0	0	0	0	0	3	2	2	2	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	